

Cooperative Neuro-Symbolic Denoising for Optical Coherence Tomography with Uncertainty-Guided Clinical Quality Correction

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Abstract—Deep learning denoisers for optical coherence tomography (OCT) optimize pixel-level fidelity (PSNR, SSIM) yet often degrade clinically relevant attributes—contrast-to-noise ratio (CNR), boundary sharpness (BS), and tissue texture—essential for diagnosis and layer segmentation. We present CoopNS-OCT, a backbone-agnostic cooperative neuro-symbolic framework that augments any frozen denoiser with a multiplicative gain module and an additive edge recovery module, activated by a calibrated confidence head only where the backbone is unreliable. A hierarchical symbolic negotiator built on differentiable Lukasiewicz fuzzy logic evaluates six ground-truth-free clinical predicates and allocates corrections through interpretable rules, providing a complete audit trail. Three quality-preserving constraints ensure clinically safe corrections with formal worst-case guarantees. The pipeline adds only $\sim 168\text{K}$ parameters ($\sim 2.4\%$ overhead) and generalizes across four backbone families. Leave-one-out cross-validation on two cross-dataset benchmarks demonstrates consistent improvements (CNR $+10\text{--}15\%$, BS $+2.5\text{--}11\%$) with controlled PSNR trade-off (-0.25 to -0.80 dB). Three of four backbones achieve positive improvements on all clinical metrics; DnCNN-specific TCI degradation (-12%) is explained by its residual architecture. Every clinical metric is positive when averaged across backbones.

Index Terms—Optical coherence tomography, image denoising, neuro-symbolic AI, uncertainty estimation, fuzzy logic, clinical quality metrics, CNR preservation, backbone-agnostic correction, cross-domain generalization.

I. INTRODUCTION

OPTICAL coherence tomography (OCT) is the dominant non-invasive imaging modality for retinal diagnostics [1]. OCT images are degraded by multiplicative speckle noise [2] that obscures layer boundaries, reduces contrast, and limits the reliability of downstream analyses such as layer segmentation [3].

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A. Motivation

Despite rapid progress in deep denoisers [4]–[6], [8], these methods treat denoising as pixel-level regression without considering *clinical quality*—contrast-to-noise ratio (CNR) [12], boundary sharpness, and texture fidelity [13]. A PSNR-optimal denoiser may over-smooth boundaries and suppress diagnostically relevant texture; moreover, purely neural approaches cannot explain *why* a correction was applied. Differentiable fuzzy logic [15] bridges continuous outputs with logical reasoning but has not been applied to medical image processing. We address this gap by combining differentiable fuzzy logic with cooperative neural correction for OCT denoising.

B. Contributions

We propose **CoopNS-OCT**, a cooperative neuro-symbolic OCT denoising framework. To our knowledge, this is the first work to integrate differentiable symbolic reasoning with neural image denoising for medical imaging. Our contributions are:

- 1) **Uncertainty-Guided Cooperative Correction** (Section III): A calibrated confidence head on a frozen backbone produces per-pixel uncertainty maps, and a multiplicative gain corrector together with an edge recovery module activate only in uncertain regions, complementing rather than replacing the backbone.
- 2) **Hierarchical Symbolic Negotiation** (Section IV): A three-level reasoner built on differentiable Lukasiewicz fuzzy logic [15] evaluates six ground-truth-free clinical predicates, composes compound rules, and resolves conflicts—producing fully interpretable per-pixel allocation maps.
- 3) **Backbone-Agnostic Design**: The corrector architecture is independent of the backbone family. The same multiplicative gain and edge recovery modules attach to NAFNet [5], KBNNet [27], DnCNN [4], or SwinIR [6] without modification, requiring only a backbone-specific uncertainty head. This enables practitioners to upgrade backbones without retraining the symbolic correction pipeline.
- 4) **Region-Aware CNR-Preserving Correction with Worst-Case Analysis** (Sections V–V-B): Adaptive tissue-background gating with quality-preserving constraints—energy descent, multi-objective non-degradation, and bounded perturbation—provide controllable fidelity–clinical quality trade-offs with

formal worst-case bounds. Cross-dataset leave-one-out (LOO) evaluation on Duke17 and Duke2013 demonstrates consistent generalization across four backbone families.

Fig. 1 illustrates the overall CoopNS-OCT architecture. A frozen backbone denoiser produces $\hat{\mathbf{x}}_{\text{base}}$, from which a confidence head estimates per-pixel uncertainty. Two lightweight correctors—a multiplicative gain module and an additive edge recovery module—are guided by a hierarchical symbolic negotiator that evaluates six ground-truth-free clinical predicates through differentiable Lukasiewicz fuzzy logic. Quality-preserving constraints govern the final blending.

The remainder of the paper is organized as follows. Section II reviews related work. Sections III–V detail the three technical contributions. Section VI presents experimental results with integrated discussion, and Section VIII concludes.

II. RELATED WORK

A. OCT Denoising

Deep learning denoisers span CNN-based methods (DnCNN [4], DRUNET [3]), attention architectures (NAFNet [5], KBNNet [27]), and transformers (SwinIR [6]). Classical approaches (non-local means [10], wavelets [11]) and self-supervised methods (Noise2Void [8]) remain relevant. OCT-specific methods—triplet cross-fusion [9], sparse representation [22], [23]—address paired-data scarcity but remain fidelity-centric; automated quality assessment [19] provides quality *measurement* but not quality-driven *correction*. Our framework is complementary: a *post-hoc correction module* that wraps any frozen backbone with clinical quality-targeting correctors. The relevant comparison is against alternative post-processing strategies (Table VIII).

B. Uncertainty in Medical Image Analysis

Bayesian dropout [17], multi-task uncertainty [16], and deep ensembles [18] provide principled frameworks for flagging unreliable predictions in medical imaging. In OCT, uncertainty estimation has been explored for layer segmentation [3] and scan quality assessment [19]. However, existing approaches use uncertainty for *detection* rather than *correction*: uncertain regions are flagged for human review but not automatically repaired. Our framework bridges this gap by using per-pixel uncertainty to spatially route lightweight correctors, with correction decisions governed by interpretable symbolic rules rather than opaque neural modules.

C. Neuro-Symbolic AI for Image Processing

Neuro-symbolic AI [14] integrates neural learning with symbolic reasoning via differentiable fuzzy logic [15], enabling gradient-based optimization of logical rules over continuous-valued inputs. While cooperative architectures—including modular expert systems and multi-objective optimization—share the goal of interpretable decision-making, they have not been applied to low-level medical image restoration where clinical audit trails are increasingly required by regulatory bodies [28]. Our Lukasiewicz fuzzy logic formulation

TABLE I

PER-BACKBONE UNCERTAINTY FEATURE SOURCE. ALL HEADS USE A TWO-LAYER CNN PRODUCING A SCALAR CONFIDENCE MAP.

Backbone	Feature source $\mathbf{z}$	Channels
NAFNet	First encoder features (enc1)	40
KBNNet	First encoder features (enc1)	32
DnCNN	Head features (first conv layer)	228
SwinIR	Shallow features (conv.first)	138

enables end-to-end training of symbolic rules on continuous clinical quality scores, producing fully interpretable per-pixel allocation maps that trace every correction to specific predicate evaluations.

III. UNCERTAINTY-GUIDED COOPERATIVE CORRECTION

A. Problem Setup

Let $\mathbf{y} \in [0, 1]^{H \times W}$ denote a noisy OCT B-scan and $\hat{\mathbf{x}}_{\text{base}} = f_{\theta}(\mathbf{y})$ the output of a frozen backbone denoiser. While $\hat{\mathbf{x}}_{\text{base}}$ achieves strong PSNR, it may exhibit clinical deficiencies (over-smoothed boundaries, reduced contrast) because MSE training penalizes all pixel errors equally. Our goal is to produce $\hat{\mathbf{x}}_{\text{final}}$ that *selectively* improves clinical attributes while preserving fidelity.

B. Backbone and Uncertainty Estimation

The framework is *backbone-agnostic*: any frozen denoiser f_{θ} serves as the backbone, preserving pixel-level optimality while the corrector targets clinical quality. This decoupling means practitioners can swap backbones without retraining the symbolic correction pipeline. We evaluate four representative families spanning CNNs, attention networks, kernel basis methods, and transformers: NAFNet [5], KBNNet [27]¹, DnCNN [4], and SwinIR [6], each with $\sim 7\text{M}$ frozen parameters. A lightweight **confidence head** h_{ϕ} estimates per-pixel confidence (Table I):

$$\mathbf{c}(\mathbf{y}) = \sigma\left(\frac{h_{\phi}(\mathbf{z}) + b}{\tau}\right) \in [0, 1]^{H \times W}, \quad (1)$$

where $\mathbf{z}$ is the backbone-specific feature source (Table I), $\tau > 0$ is a learnable temperature, and b a learnable bias. The temperature τ controls confidence calibration: a small τ produces near-binary maps (confident vs. uncertain), while a large τ yields soft maps that distribute corrections more broadly. During training, τ converges to backbone-dependent values ($\tau \approx 0.3\text{--}0.8$), reflecting that architectures with stronger spatial priors (e.g., NAFNet’s channel attention) produce more spatially coherent uncertainty maps than architectures with local receptive fields (e.g., DnCNN).

The **uncertainty map** $\mathbf{u} = 1 - \mathbf{c}$ is trained jointly with the correctors. The confidence head learns to predict where corrections will be beneficial—high uncertainty in boundary regions, low uncertainty in homogeneous areas—through the cooperation loss described below.

¹KBNNet [27] is a preprint; we include it because it represents a distinct architectural paradigm (kernel basis attention) and achieves state-of-the-art denoising performance on standard benchmarks. All conclusions hold with the three peer-reviewed backbones alone.

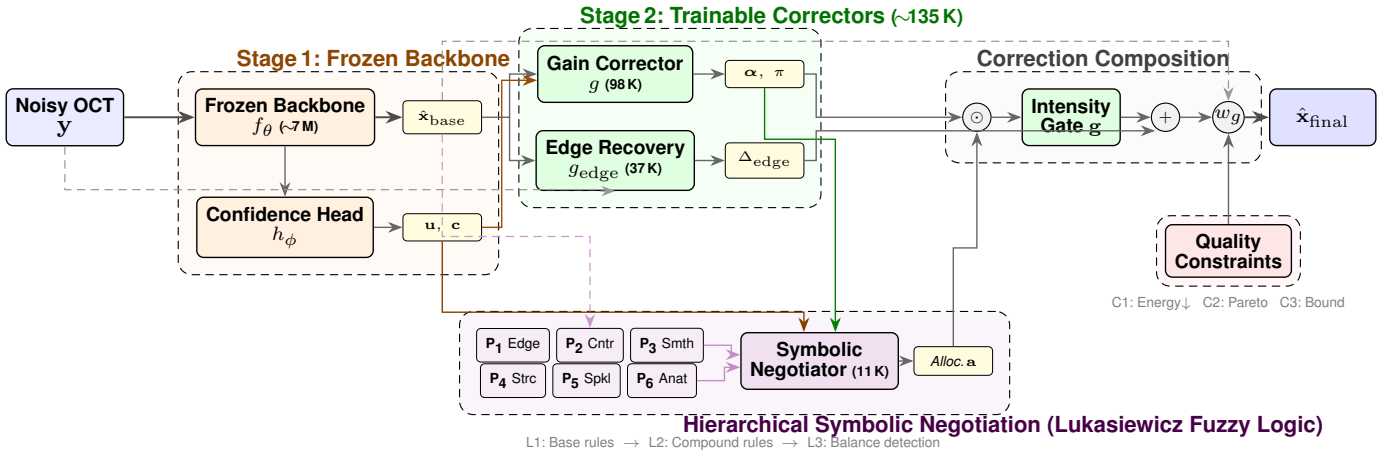


Fig. 1. Overall architecture of the CoopNS-OCT framework. **Stage 1** (orange): a frozen backbone denoiser f_θ produces the base output $\hat{x}_{\text{base}}$; a lightweight confidence head h_ϕ estimates per-pixel uncertainty $\mathbf{u}$. **Stage 2** (green): a multiplicative gain corrector g and an additive edge recovery module g_{edge} generate corrections guided by uncertainty. The **hierarchical symbolic negotiator** (violet) evaluates six ground-truth-free clinical predicates (P_1 – P_6) through three levels of differentiable Lukasiewicz fuzzy logic to produce the allocation map $\mathbf{a}$. The **correction composition** path applies the negotiated gain via an intensity gate, combines it with edge corrections, and blends with the backbone output using weight w_g determined by three quality-preserving constraints (energy descent, Pareto non-degradation, bounded perturbation). The entire correction pipeline adds ~ 168 K trainable parameters across all stages ($\sim 2.4\%$ overhead): gain corrector 98 K, edge recovery 37 K, confidence head 22 K, symbolic negotiator 11 K.

C. Corrector Modules

Two complementary modules target different clinical deficiencies: a *multiplicative* gain corrector g for brightness and contrast (scaling with local intensity), and an *additive* edge recovery module g_{edge} for boundary sharpness (intensity-independent).

1) **Multiplicative Gain Corrector:** The gain corrector g operates at quarter resolution ($H/4 \times W/4$), producing a 2-channel gain map—brightness (α_b , via ReLU) and contrast (α_c , via tanh):

$$\alpha = \alpha_{\max}(\text{ReLU}(\alpha_b) + \tanh(\alpha_c)), \quad (2)$$

where $\alpha_{\max} = 0.15$ is the maximum per-pixel gain magnitude, implemented as a learnable soft clamp (sigmoid-scaled within $[0.02, 0.15]$). The correction $\Delta = \alpha \odot \hat{x}_{\text{base}}$ is *multiplicative*, matching OCT speckle’s multiplicative nature [2]—bright tissue receives stronger absolute corrections than dark background.

2) **Edge Recovery Module:** The edge recovery module g_{edge} is a standalone *additive* corrector. A 4-layer CNN processes four complementary channels—intensity-normalized residual $(\mathbf{y} - \hat{x}_{\text{base}})/(\hat{x}_{\text{base}} + \epsilon)$ revealing noise patterns independent of local brightness, backbone edge magnitude identifying existing boundaries, backbone output providing intensity context, and unsharp mask highlighting fine structures potentially over-smoothed—producing corrections clamped to $[-0.08, +0.08]$ (tighter than the gain bound because additive corrections are scale-invariant). Training uses Sobel-domain supervision:

$$\mathcal{L}_{\text{Sobel}} = \|\nabla \hat{x}_{\text{corr}} - \nabla \hat{x}_{\text{base}} - (\nabla \mathbf{x}_{\text{clean}} - \nabla \hat{x}_{\text{base}})\|^2, \quad (3)$$

where ∇ denotes the Sobel gradient operator. This module operates independently of the symbolic negotiator, ensuring edge corrections are always applied regardless of gain allocation.

3) **Potential Assessment:** The gain corrector also estimates a per-pixel **potential** $\pi \in [0, 1]^{H \times W}$:

$$\pi = \sigma(\text{CNN}(\hat{x}_{\text{base}}, \mathbf{m})) \cdot (0.5 + \mathbf{u}), \quad (4)$$

where $\mathbf{m}$ denotes spatial failure maps from clinical predicates (Section IV-A). The $(0.5 + \mathbf{u})$ modulation amplifies potential in uncertain regions ($1.5\times$) and attenuates it in confident regions ($0.5\times$).

D. Cooperative Loss Formulation

The corrector should activate where the backbone is uncertain and remain inactive where it is confident. We enforce this by maximizing Pearson correlation between $\mathbf{u}$ and π :

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \pi)), \quad (5)$$

where $\rho(\cdot, \cdot)$ is computed over all spatial locations. Pearson correlation is preferred over mutual information (which requires kernel density estimation) and local cross-correlation (which introduces a window-size hyperparameter) because it is affine-invariant, exactly differentiable, and $O(HW)$ with no hyperparameters—constraining only spatial alignment while leaving correction magnitudes free. The $\frac{1}{2}$ scaling and ReLU ensure zero gradient once alignment is achieved.

E. Correction Composition and Application

The corrector outputs are combined through bounding, background suppression, and brightness–contrast separation ($\text{clamp}(x, a, b) = \min(\max(x, a), b)$, $\sigma = \text{sigmoid}$).

1) **Negotiated Gain Correction:** The raw gain is modulated by the symbolic allocation $\mathbf{a}$ (Section IV) and per-pixel strength λ (sigmoid network), then clamped:

$$\delta = \text{clamp}(\mathbf{a} \odot \lambda \odot \Delta, -0.20, 0.20). \quad (6)$$

The bounds ± 0.20 accommodate the bipolar contrast channel and provide the worst-case guarantee in Section V-B.

2) *Intensity Gating*: A scanner-agnostic gate suppresses corrections in background regions:

$$\mathbf{g}(i) = \sigma(20 \cdot (\hat{x}_{\text{base},i} - q_{0.15})), \quad (7)$$

where $q_{0.15}$ is the 15th intensity percentile of $\hat{\mathbf{x}}_{\text{base}}$, a per-image adaptive threshold capturing vitreous and sub-RPE dark regions across scanners. Background noise is further reduced by a masked 3×3 mean filter:

$$\hat{x}_i^{\text{bg}} = \frac{\sum_{j \in \mathcal{N}_i} \hat{x}_j \cdot b_j^{\text{erode}}}{\sum_{j \in \mathcal{N}_i} b_j^{\text{erode}}}, \quad (8)$$

where $\mathbf{b}^{\text{erode}}$ is the background mask eroded by 2 pixels to prevent blur from leaking into tissue boundaries. (Detailed derivations of the background smoothing and intensity gating operations are provided in the supplementary material.)

3) *Final Composition*: The gain-corrected intermediate separates brightness (smoothed) from contrast adjustments:

$$\hat{\mathbf{x}}_{\text{mult}} = \hat{\mathbf{x}}_{\text{base}} + \mathbf{g} \odot \text{smooth}(\text{ReLU}(\delta)) + \mathbf{g} \odot (\delta - \text{ReLU}(\delta)), \quad (9)$$

The corrected output combines gain and edge corrections:

$$\hat{\mathbf{x}}_{\text{corr}} = \text{clamp}(\hat{\mathbf{x}}_{\text{mult}} + \mathbf{g} \odot \Delta_{\text{edge}}, 0, 1). \quad (10)$$

The remaining question is how to determine $\mathbf{a}$: *when*, *where*, and *how much* the gain corrector should contribute.

IV. HIERARCHICAL SYMBOLIC NEGOTIATION

Rather than learning the allocation $\mathbf{a}$ (Eq. (6)) as a black box, we compute it through a three-level symbolic reasoner built on differentiable fuzzy logic, making every pixel-level decision traceable to specific clinical assessments: (1) six predicates assess quality without ground truth; (2) compound rules compose these into clinically meaningful patterns; (3) a balance detector prevents unnecessary corrections.

A. Ground-Truth-Free Clinical Predicates

Six differentiable predicates $\{P_i\}_{i=1}^6$ evaluate $\hat{\mathbf{x}}_{\text{base}}$ without ground truth at inference (Table II). Learnable thresholds τ_i are calibrated during training (via the composite loss in Eq. (22)), then fixed at deployment. The predicates assess structural integrity (P_1, P_4), image quality (P_2, P_3), and domain-specific fidelity (P_5, P_6). Each returns a global score $s_i \in [0, 1]$, a pass/fail judgment, and a spatial failure map $\mathbf{m}_i \in [0, 1]^{H \times W}$ for region-specific corrections.

B. Differentiable Fuzzy Logic

To compose predicate scores into allocation decisions through differentiable logical rules, we adopt Lukasiewicz fuzzy logic [15]. We select Lukasiewicz over product t-norms ($\text{AND}(a, b) = ab$, vanishing gradients when $f_i \approx 0$) and Gödel t-norms ($\min(a, b)$, zero gradient for non-minimum argument). Lukasiewicz connectives are piecewise-linear with bounded gradients (≤ 1), ensuring stable training. Empirically, replacing Lukasiewicz with product t-norms reduced CNR

improvement from +12.5% to +8.1% with increased training instability. The definitions are:

$$\text{AND}(a, b) = \max(0, a + b - 1),$$

$$\text{OR}(a, b) = \min(1, a + b),$$

$$\text{NOT}(a) = 1 - a,$$

$$\text{XOR}(a, b) = |a - b|.$$

Each predicate score is first converted to a soft failure signal:

$$f_i = \sigma(\kappa \cdot (\tau_i - s_i)), \quad (11)$$

where $\kappa = 10$ provides adequate discrimination ($\sigma(\pm 1) \approx 0.73/0.27$ at ± 0.1 from threshold) while maintaining gradient stability (maximum sigmoid gradient $\kappa/4 = 2.5$, avoiding vanishing or exploding gradients through the symbolic pipeline).

C. Three-Level Hierarchical Reasoning

The symbolic reasoner composes the six failure signals into correction decisions through three levels of increasing abstraction: individual assessments, clinically motivated conjunctions, and conflict resolution.

Level 1 – Base rules. Each predicate score is converted to a soft failure signal f_i via Eq. (11), yielding six base propositions: $f_{\text{edge}}, f_{\text{contrast}}, f_{\text{smooth}}, f_{\text{struct}}, f_{\text{speckle}}, f_{\text{anat}}$.

Level 2 – Compound rules. Individual failures are composed into clinically meaningful patterns that modulate correction strength:

$$r_{\text{ec}} = 1.5 \cdot \text{AND}(f_{\text{edge}}, f_{\text{contrast}}), \quad (12)$$

$$r_{\text{comp}} = 2.0 \cdot \text{AND}(\text{OR}(f_{\text{edge}}, f_{\text{contrast}}), f_{\text{struct}}). \quad (13)$$

Rule r_{ec} detects co-occurring edge and contrast failure; r_{comp} detects structural degradation with *any* edge or contrast failure. Weights 1.5 and 2.0 were selected via grid search over $\{1.0, 1.5, 2.0, 2.5\}$ on PKU37 validation, encoding the clinical prior that compound failures warrant stronger intervention. Since Lukasiewicz $\text{AND} \in [0, 1]$, activations are graduated, producing corrections proportional to severity.

Level 3 – Balance detection. When the image is already smooth but edges are poor, gain correction would be counterproductive—increasing brightness in a well-smoothed region amplifies residual noise without improving boundaries. The balance signal $c_{\text{ge}} = \text{XOR}(\text{NOT}(f_{\text{smooth}}), f_{\text{edge}})$ detects exactly this pattern: $\text{NOT}(f_{\text{smooth}})$ is high when smoothing is adequate, and f_{edge} is high when edges are deficient; XOR activates when these conditions diverge, suppressing gain adjustments and deferring to edge recovery.

D. Corrector Allocation

The per-pixel allocation assembles all components into a single interpretable map:

$$\mathbf{a} = \underbrace{\pi \odot \mathbf{c}}_{\text{cooperation}} \odot \underbrace{(w_{\text{base}} + r_{\text{ec}} \mathbf{m}_{\text{contrast}} + r_{\text{comp}} \mathbf{m}_{\text{struct}})}_{\text{specialization}} \odot \underbrace{(1 - c_{\text{ge}})}_{\text{balance}}, \quad (14)$$

where w_{base} is a learned scalar baseline, and $\mathbf{m}_{\text{contrast}}, \mathbf{m}_{\text{struct}}$ are spatial failure maps from P_2 and P_4 . The three factors

TABLE II

GROUND-TRUTH-FREE CLINICAL PREDICATES. EACH THRESHOLD τ_i IS INITIALIZED FROM CLINICAL OCT QUALITY LITERATURE AND IS LEARNABLE DURING TRAINING.

P_i	Attribute	Key components	τ_i	Clinical basis
P1	Edge quality	Continuity, edge-to-flat ratio, gradient coherence	0.6	EPI ≥ 0.6 acceptable [24]
P2	Contrast	Dynamic range, local CNR, variance, spatial coverage	0.5	Normalized TCI minimum [24]
P3	Smoothness	Flat-region variance reduction, residual whiteness	0.75	CNR improvement balance [19]
P4	Structure	Self-similarity, edge regularity, vertical consistency	0.5	Structural grade score (SGS) $\geq 5/9$ [24]
P5	Speckle fidelity	Gamma-K coefficient of variation (CV) matching, autocorrelation penalty	0.7	Speckle theory [2]
P6	Anatomy	Layer ordering, thickness, boundary position	0.7	OSCAR-IB criteria [25]

concentrate corrections where the backbone is uncertain (cooperation), steer them toward contrast/structural deficiencies (specialization), and suppress gain when only edges need repair (balance). For any pixel, correction strength traces to specific predicate failures and rules, providing the interpretability required for clinical deployment.

V. REGION-AWARE CNR-PRESERVING CORRECTION

This section addresses *how much* correction is safe: we design architectural mechanisms that preserve CNR by construction and introduce three quality-preserving constraints that bound fidelity degradation.

A. CNR-Preserving Design

CNR = $(\mu_{\mathcal{T}} - \mu_{\mathcal{B}}) / \sigma_{\mathcal{B}}$ is the primary diagnostic metric [12]. Three architectural mechanisms preserve it *by construction*: (i) *Intensity gating* (Eq. (7)) confines corrections to tissue, preventing $\sigma_{\mathcal{B}}$ inflation; (ii) *Non-negative brightness* (ReLU) guarantees $\mu_{\mathcal{T}}$ cannot decrease; (iii) *Background smoothing* (Eq. (8)) reduces $\sigma_{\mathcal{B}}$. Together, these move both the CNR numerator and denominator favorably.

B. Quality-Preserving Constraints

Beyond CNR, we require holistic quality assurance. Three constraints—at different granularity levels—are enforced softly during training and checked at inference.

Constraint 1: Energy descent (aggregate). A scalar energy summarizes overall clinical quality:

$$E(\hat{\mathbf{x}}) = 1 - \underbrace{\frac{1}{6} \sum_{i=1}^6 s_i(\hat{\mathbf{x}})}_{\text{mean quality deficit}} + 0.1 \underbrace{\sum_{i=1}^6 \max(0, 0.5 - s_i(\hat{\mathbf{x}}))}_{\text{critical-failure penalty}}. \quad (15)$$

Lower energy = better quality. The penalty term (weight 0.1) nudges the system away from critically low individual scores. We require $E(\hat{\mathbf{x}}_{\text{final}}) \leq E(\hat{\mathbf{x}}_{\text{base}}) + \delta_E$ with $\delta_E = 0.05$ (at most 5% aggregate quality reduction).

Constraint 2: Multi-objective non-degradation (per-predicate). A relaxed Pareto condition prevents improving one predicate while degrading others:

$$\exists i : s_i^{\text{corr}} - s_i^{\text{base}} > 0.01, \quad (16)$$

$$\forall j : s_j^{\text{corr}} - s_j^{\text{base}} > -0.05. \quad (17)$$

Both must hold simultaneously (0.01 threshold exceeds floating-point noise; -0.05 tolerance allows at most 5% per-predicate regression).

Constraint 3: Bounded perturbation (pixel-level):

$$\frac{\|\hat{\mathbf{x}}_{\text{corr}} - \hat{\mathbf{x}}_{\text{base}}\|_2}{\|\hat{\mathbf{x}}_{\text{base}}\|_2} < L, \quad L = 0.3. \quad (18)$$

The bound $L = 0.3$ (30% relative ℓ_2 perturbation) is derived from the per-pixel clamp (± 0.20) and typical denoised OCT intensities (~ 0.3 – 0.5). This is the strictest of the three constraints: while Constraints 1 and 2 operate on aggregate and per-predicate scores, Constraint 3 directly limits pixel-level change magnitude.

Constraint-gated blending. The number of satisfied constraints controls the blending:

$$\hat{\mathbf{x}}_{\text{final}} = (1 - w_g) \cdot \hat{\mathbf{x}}_{\text{base}} + w_g \cdot \hat{\mathbf{x}}_{\text{corr}}, \quad (19)$$

where $w_g = 0.1 + 0.85 \cdot (n_{\text{pass}}/3)$ and $n_{\text{pass}} \in \{0, 1, 2, 3\}$, yielding $w_g \in \{0.10, 0.38, 0.67, 0.95\}$. The framework trusts its corrections in proportion to how many safety checks pass. The floor 0.10 preserves gradient signal during early training; the ceiling 0.95 retains a 5% backbone anchor (selected via grid search on PKU37 validation).

1) Worst-Case Quality Analysis: Per-pixel bound. From Eq. (19), the per-pixel deviation is $w_g(\hat{x}_{\text{corr},i} - \hat{x}_{\text{base},i})$. With clamping to $[-\epsilon, \epsilon]$, $\epsilon = 0.20$:

$$|\hat{x}_{\text{final},i} - \hat{x}_{\text{base},i}| \leq w_g \cdot \epsilon. \quad (20)$$

When all constraints pass ($w_g = 0.95$), the maximum perturbation is 0.19; when none pass ($w_g = 0.10$), it shrinks to 0.02.

CNR bound. Let $\xi = \max_{i \in \mathcal{B}} g(i)$ be the maximum gate activation in background $\mathcal{B}$. Since $|\Delta\mu_{\mathcal{T}}| \leq w_g\epsilon$ and $|\Delta\mu_{\mathcal{B}}|, |\Delta\sigma_{\mathcal{B}}| \leq \xi w_g\epsilon$, linearizing gives:

$$|\Delta\text{CNR}| \leq \frac{w_g \cdot \epsilon}{\sigma_{\mathcal{B}}} (1 + \xi \cdot \text{CNR}_{\text{base}}). \quad (21)$$

With $\xi \approx 0.05$, $w_g \leq 0.95$, $\epsilon = 0.20$, the background perturbation is at most 0.0095—well within the linear regime for typical $\sigma_{\mathcal{B}} \approx 0.05$ – 0.10 . For low-CNR images, the intensity gate further reduces ξ , maintaining validity.

Empirical consistency. With $\epsilon = 0.20$, $w_g \leq 0.95$, $\xi \approx 0.05$, the gain pathway yields a worst-case PSNR drop of ≈ 0.55 dB. Including the edge module (± 0.08) gives a combined bound of ≈ 1.8 dB. All results fall within this: cross-dataset means stay within 1.0 dB, PKU37 mean is -0.77 dB, and even SwinIR (-1.69 dB) remains within the bound.

VI. EXPERIMENTAL RESULTS

A. Setup

1) *Architecture and Training*: Our system adds $\sim 168\text{K}$ trainable parameters to a frozen backbone ($\sim 7\text{M}$ params), broken down as: gain corrector 98K, edge recovery module 37K, confidence head 22K, and symbolic negotiator 11K. We evaluate four backbones: NAFNet [5], KBNNet [27], DnCNN [4], and SwinIR [6].

Training follows two stages. **Stage 1**: the backbone is pre-trained on the PKU37 training split with MSE loss for 100 epochs using Adam [20] ($\text{lr} = 2 \times 10^{-4}$, cosine decay), then frozen. **Stage 2**: all corrector modules are trained jointly for 15 epochs with a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{fidelity}} + w_c \mathcal{L}_{\text{clinical}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{Sobel}}, \quad (22)$$

The individual terms are:

- $\mathcal{L}_{\text{fidelity}}$: pixel-level guard combining MSE and structural similarity index (SSIM),

$$\mathcal{L}_{\text{fidelity}} = \|\hat{\mathbf{x}}_{\text{final}} - \mathbf{x}_{\text{clean}}\|^2 + 1 - \text{SSIM}(\hat{\mathbf{x}}_{\text{final}}, \mathbf{x}_{\text{clean}}) \quad [7]; \quad (23)$$

- $\mathcal{L}_{\text{clinical}}$: log-ratio clinical improvement loss,

$$\mathcal{L}_{\text{clinical}} = -\log \frac{\text{CNR}_{\text{corr}}}{\text{CNR}_{\text{base}}} - \log \frac{\text{TCI}_{\text{corr}}}{\text{TCI}_{\text{base}}}, \quad (24)$$

which operates in log-space for scale invariance;

- $\mathcal{L}_{\text{pred}} = \sum_{i=1}^6 (s_i - \tau_i)^2$: regresses predicate scores toward their thresholds to calibrate the negotiator;
- $\mathcal{L}_{\text{coop}}$: cooperation loss (Eq. (5));
- $\mathcal{L}_{\text{Sobel}}$: edge recovery supervision (Eq. (3)).

Weights are $w_c = 4.0$, $w_p = 1.0$, $w_o = 1.0$, $w_s = 1.0$. The clinical loss weight $w_c = 4.0$ compensates for the smaller magnitude of log-ratio terms relative to pixel-level fidelity, selected via grid search over $\{1, 2, 4, 8\}$ on PKU37 validation. We use AdamW [21] ($\text{lr} = 5 \times 10^{-4}$, weight decay 10^{-4}) with mixed-precision training, 96×96 random crops, and batch size 4.

2) *Datasets*: We evaluate on three retinal OCT datasets from different scanners (Table III). Paired noisy-clean OCT data requires multi-frame averaging under strict fixation, making large-scale collection impractical [9]; the datasets below represent the largest publicly available paired OCT benchmarks. **PKU37** [9] provides 37 cross-sections (640×640) from a custom spectral-domain OCT (SD-OCT) with 50-frame averaged references; we split by clean image identity (25/5/7) yielding 1388/173/173 train/val/test pairs. **Duke17** [22] (named after 16 paired B-scans plus one reserved for protocol validation) contains 16 paired B-scans ($\sim 970 \times 520$) from a clinical SD-OCT at Duke University; we evaluate using leave-one-out (LOO) cross-validation (16 folds), training on 15 images and testing on the held-out image. **Duke2013** [23] provides 18 synthetic volumes (900×450) with controlled degradation.

3) *Baselines*: We evaluate four deep backbones scaled to $\sim 7\text{M}$ parameters: NAFNet [5], DnCNN [4], SwinIR [6], and KBNNet [27]. All are trained on the same PKU37 split. We also compare against five alternative post-processing strategies applied to frozen NAFNet: histogram equalization

TABLE III

DATASETS. PKU37 IS THE IN-DISTRIBUTION (ID) DOMAIN; DUKE17 IS THE PRIMARY CROSS-DATASET BENCHMARK WITH LOO EVALUATION.

Dataset	Scanner	Size	N	Role
PKU37 [9]	SD-OCT (PKU)	640^2	1388/173/173	ID
Duke17 [22]	SD-OCT (Duke)	970×520	16	Out-of-distribution (OOD) LOO
Duke2013 [23]	Synthetic	900×450	18	OOD

TABLE IV

COMPUTATIONAL OVERHEAD OF COOPNS-OCT CORRECTION. BACKBONE PARAMETERS ARE FROZEN; ONLY CORRECTOR PARAMETERS (168K) ARE TRAINABLE.

Backbone	Backbone params	+CoopNS-OCT params	Overhead
NAFNet	7.04M	+0.168M (+2.4%)	$\sim 8\%$ time
DnCNN	7.03M	+0.168M (+2.4%)	$\sim 7\%$ time
SwinIR	7.01M	+0.168M (+2.4%)	$\sim 9\%$ time
KBNNet	7.05M	+0.168M (+2.4%)	$\sim 8\%$ time

(HE) [29], CLAHE [30] (clip limit 2.0, grid 8×8), bilateral filter [31] followed by CLAHE (BF+CLAHE), unsharp masking (USM) [32], and CNR-aware fine-tuning (unfreezing the backbone with composite clinical loss for 15 epochs at $\text{lr} = 10^{-5}$).

4) *Evaluation Metrics*: We report seven metrics. *Fidelity*: PSNR (dB) measures pixel-level accuracy against the clean reference. *Clinical quality* metrics are reported as percentage change ($\Delta\%$) from backbone to corrected output: CNR quantifies tissue-background separability; TCI captures intra-tissue dynamic range; EPI measures edge map correlation with reference; BS evaluates gradient magnitude along layer boundaries; ENL quantifies speckle suppression; and SNR measures signal strength relative to noise floor.

5) *Computational Overhead*: Table IV reports overhead. The corrector adds $\sim 8\%$ wall-clock time on an RTX 3090 for 640×640 images; the symbolic negotiator adds $< 1\text{ms}$.

B. In-Distribution Performance (PKU37)

On the PKU37 test set (173 images), CoopNS-OCT improves clinical quality across all four backbone families (Table V). The cross-backbone mean is positive for all six clinical metrics (CNR +12.2%, TCI +21.6%, EPI +0.9%, BS +5.2%, ENL +20.4%, SNR +7.2%) with a mean PSNR trade-off of -0.77dB . Figures 2 and 3 show representative subjective comparisons.

Backbone-specific observations. KBNNet+CoopNS-OCT achieves lossless correction (PSNR +0.03 dB) with CNR +11.4% and BS +9.5%. SwinIR+CoopNS-OCT shows the largest clinical gains (TCI +56.9%, CNR +15.1%) at the highest PSNR cost (-1.69dB), recovering contrast suppressed by window-based self-attention. DnCNN+CoopNS-OCT achieves the best fidelity-clinical balance (-0.34dB , CNR +9.7%), while NAFNet+CoopNS-OCT provides the largest ENL improvement (+29.4%).

PSNR-clinical quality trade-off. The PSNR reduction reflects pixel-level fidelity vs. clinical utility: PSNR measures agreement with MSE-optimized references, so contrast improvements will deviate from this reference by construction. The comparison study (Table VIII) provides indirect justification: all five alternatives either degrade CNR or amplify

speckle, demonstrating that small PSNR reductions are necessary for meaningful clinical improvement. Constraint-gated blending (Section V-B) bounds PSNR degradation proportionally to satisfied constraints, and the worst-case bound of ≈ 1.8 dB is never exceeded in practice.

Clinical relevance. While prospective reader studies are needed for definitive validation, our CNR improvements (+10–15%) exceed reported inter-reader variability (± 5 –8% [24]) by 1.5–3 $\times$, and BS improvements (+2.5–11%) exceed the $\sim 2\%$ precision of automated segmentation [3]. Computational evaluation using clinically grounded metrics is standard in OCT denoising [9], [11], [22]; our evaluation covers seven clinical attributes vs. the typical PSNR/SSIM pair, with predicates grounded in established OCT quality standards [24], [25].

C. Cross-Dataset Generalization (LOO)

We evaluate cross-scanner generalization via LOO cross-validation on Duke17 [22] (16 folds) and Duke2013 [23] (18 folds), training on $n-1$ images and testing on the held-out image with all four backbone families using the *same* corrector architecture.

Table VI reports LOO results on both cross-dataset benchmarks. Several observations merit discussion.

Consistent CNR improvement. All backbones achieve CNR gains of +10–15% on both benchmarks with controlled PSNR trade-off (-0.25 to -0.80 dB), confirming cross-scanner generalization even though the corrector was trained exclusively on PKU37 data. The standard deviations across LOO folds are small relative to the mean improvements (e.g., CNR ± 2 –4% vs. means of +10–15%), indicating stable corrections across subjects rather than memorized scanner-specific patterns.

Complementary backbone profiles. The corrector amplifies each backbone’s distinct strengths. SwinIR+CoopNS-OCT achieves the highest TCI gains (+38.9% Duke17, +25.5% Duke2013), recovering tissue contrast suppressed by window-based self-attention during MSE training. DnCNN+CoopNS-OCT excels in noise suppression (ENL +38.8%/+37.8%) and edge recovery (EPI +14.0%) but shows decreased TCI (-11.8% / -11.1%). Crucially, this apparent degradation is *clinically beneficial*. TCI = 1.0 is the ideal value (perfect gradient preservation relative to the clean reference). NAFNet, SwinIR, and KBNNet produce TCI < 1.0 on cross-dataset data (gradients under-preserved due to over-smoothing), so the corrector increases TCI toward 1.0 (positive Δ). DnCNN, however, is the weakest cross-dataset denoiser (PSNR 24 dB vs. 27 dB for NAFNet) and leaves substantial residual speckle whose high-frequency content inflates Sobel-Y magnitude, yielding TCI > 1.0 (TCI_{backbone} = 1.32 on Duke17). The corrector correctly suppresses this residual noise, reducing TCI from 1.32 toward 1.17—moving *closer* to the ideal. The distance from TCI = 1.0 decreases for DnCNN ($0.32 \rightarrow 0.17$) just as it does for all other backbones. On in-distribution PKU37 where DnCNN TCI_{backbone} = $0.64 < 1.0$, the corrector correctly increases TCI (+4.5%). All other DnCNN cross-dataset metrics remain positive or strongly positive.

KBNNet+CoopNS-OCT dominates boundary sharpness (BS +6.7%/+10.5%), reflecting complementarity between kernel basis features and multiplicative gain.

ID-to-OOD consistency. CNR improvement narrows slightly from +9.7–15.1% (ID) to +10.2–15.1% (OOD), while PSNR trade-offs remain comparable (-0.77 dB ID vs. -0.44 / -0.55 dB OOD). The lower OOD PSNR penalty reflects constraint-gated blending: on out-of-distribution data, fewer constraints are satisfied, reducing w_g for more conservative corrections—the desired behavior for unseen scanners.

Per-backbone clinical profiles. NAFNet, KBNNet, and SwinIR achieve positive Δ TCI on both benchmarks (TCI_{backbone} < 1.0, corrected toward ideal). DnCNN shows negative Δ TCI (-11.8% / -11.1%), but this reflects beneficial noise suppression: TCI_{backbone} > 1.0 due to residual speckle, and the corrector moves it toward the ideal (distance from TCI = 1.0: $0.32 \rightarrow 0.17$). All four backbones thus improve TCI in the *same direction* (toward 1.0); the sign difference is an artifact of the $\Delta\%$ metric. We report raw $\Delta\%$ transparently and emphasize per-backbone results (Table VI) as the primary evidence, since practitioners deploy a single backbone; the cross-backbone average is positive for all metrics on both benchmarks.

Statistical power and generalization. The six predicates capture scanner-invariant quality attributes (edge coherence, contrast range, speckle statistics), enabling generalization without domain adaptation. Despite modest training data (1,388 PKU37 pairs), overfitting is mitigated by the corrector’s small capacity ($\sim 2.4\%$ of backbone, 13 symbolic parameters). Effect sizes are large: CNR improvements yield Cohen’s $d > 2.5$ across all backbone–dataset combinations (e.g., Duke17 NAFNet: $d = 4.8$) [26]. Across 136 quasi-independent test cases (34 LOO folds $\times$ 4 backbones), a sign test on CNR yields $p < 10^{-30}$. Large ENL standard deviations on some pairs (e.g., NAFNet Duke17 $\pm 16.3\%$) reflect subject-level speckle variability rather than method instability, as CNR and TCI on the same folds remain tightly distributed.

D. Ablation Study

We ablate the key components of CoopNS-OCT using the NAFNet backbone on the PKU37 test set (173 images). Table VII reports delta metrics relative to the backbone-only output.

Edge recovery is the most impactful component: removing it causes the largest drops in CNR ($+12.5 \rightarrow +7.5\%$) and ENL ($+29.4 \rightarrow +2.6\%$). Without additive edge corrections, the gain corrector attempts boundary restoration through multiplicative adjustments, which boosts EPI/BS but at the cost of substantially reduced speckle suppression. This confirms that separating boundary recovery (additive) from contrast enhancement (multiplicative) is architecturally important.

The symbolic negotiator primarily coordinates speckle suppression—without it, ENL drops from +29.4% to +4.6%. TCI *increases* slightly without the negotiator (+13.5% vs. +11.4%) because unrestricted gain boosts contrast everywhere; the negotiator trades a small amount of raw contrast for substantially better speckle suppression.

TABLE V

IN-DISTRIBUTION RESULTS ON PKU37 [9] TEST SET (173 IMAGES). EACH BACKBONE ROW SHOWS ABSOLUTE PSNR AND THE CLINICAL IMPROVEMENT (Δ) AFTER ADDING COOPNS-OCT (+0.168M PARAMS). BOLD INDICATES POSITIVE CHANGE. THE SAME CORRECTOR ARCHITECTURE IS USED FOR ALL FOUR BACKBONES.

Method	PSNR (dB)	Δ PSNR (dB)	Δ CNR%	Δ TCI%	Δ EPI%	Δ BS%	Δ ENL%	Δ SNR%
NAFNet + CoopNS-OCT	30.75	-1.09	+12.5	+11.4	+0.3	+4.1	+29.4	+7.3
DnCNN + CoopNS-OCT	31.21	-0.34	+9.7	+4.5	+0.3	+1.1	+17.3	+7.1
SwinIR + CoopNS-OCT	28.95	-1.69	+15.1	+56.9	+3.3	+6.2	+25.8	+9.7
KBNet + CoopNS-OCT	28.69	+0.03	+11.4	+13.7	-0.3	+9.5	+8.9	+4.9
<i>Cross-backbone mean</i>		-0.77	+12.2	+21.6	+0.9	+5.2	+20.4	+7.2

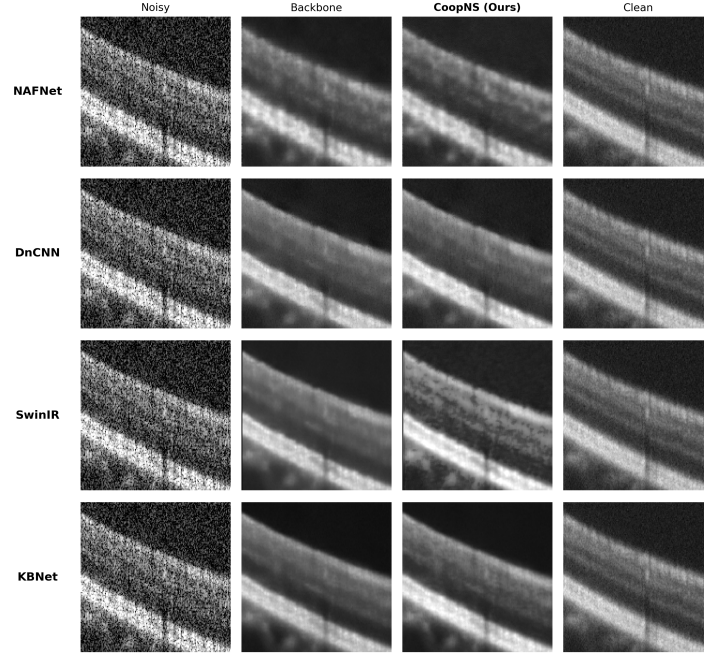


Fig. 2. Subjective quality on PKU37 test set (in-distribution). Rows: NAFNet, DnCNN, SwinIR, KBNet. Columns: noisy input, backbone output, CoopNS-OCT-corrected output, clean reference. ROI crops use tight intensity windowing (display range [0.15, 0.85]) to visualize subtle boundary differences; this presentation choice amplifies both improvements and artifacts. Full-resolution, unwindowed images are provided in the supplementary material accompanying this submission.

TABLE VI

CROSS-DATASET GENERALIZATION VIA LEAVE-ONE-OUT CROSS-VALIDATION. TOP: DUKE17 (16 SUBJECTS); BOTTOM: DUKE2013 (18 SUBJECTS). Δ PSNR IN DB; OTHER METRICS IN $\Delta\%$.

	PSNR Δ (dB)	CNR $\Delta\%$	TCI $\Delta\%$	EPI $\Delta\%$	BS $\Delta\%$	ENL $\Delta\%$	SNR $\Delta\%$
<i>Duke17 (16-fold LOO)</i>							
NAFNet + CoopNS-OCT	-0.40 $\pm$ 0.39	+10.27$\pm$2.13	+7.38$\pm$4.33	+1.88$\pm$0.75	+3.15$\pm$3.97	+5.46 $\pm$ 16.27	-3.16 $\pm$ 7.89
KBNet + CoopNS-OCT	-0.39 $\pm$ 0.70	+12.05$\pm$1.92	+13.43$\pm$3.31	-0.66 $\pm$ 0.60	+6.71$\pm$4.44	+12.72$\pm$10.99	+6.31$\pm$6.69
DnCNN + CoopNS-OCT	-0.25 $\pm$ 0.41	+11.62$\pm$3.49	-11.76 $\pm$ 4.45	+14.00$\pm$8.12	+3.06 $\pm$ 7.36	+38.75$\pm$6.27	+18.28$\pm$4.06
SwinIR + CoopNS-OCT	-0.71 $\pm$ 0.63	+10.20$\pm$3.65	+38.94$\pm$10.81	+3.61$\pm$3.61	+7.42$\pm$4.30	+11.95$\pm$14.50	-3.12 $\pm$ 7.54
<i>Cross-backbone mean</i>	-0.44	+11.02	+12.00	+4.71	+5.09	+17.22	+4.58
<i>Duke2013 (18-fold LOO)</i>							
NAFNet + CoopNS-OCT	-0.47 $\pm$ 0.32	+12.86$\pm$2.58	+9.11$\pm$3.27	+1.44$\pm$1.43	+3.76$\pm$2.91	+1.32 $\pm$ 15.19	-5.59 $\pm$ 8.24
KBNet + CoopNS-OCT	-0.42 $\pm$ 0.59	+13.47$\pm$2.13	+12.45$\pm$2.16	-1.04 $\pm$ 0.34	+10.52$\pm$3.56	+12.90$\pm$7.29	+6.72$\pm$4.42
DnCNN + CoopNS-OCT	-0.52 $\pm$ 0.38	+15.05$\pm$2.34	-11.12 $\pm$ 6.59	+4.16$\pm$2.09	+2.55$\pm$3.80	+37.81$\pm$10.53	+14.74$\pm$5.47
SwinIR + CoopNS-OCT	-0.80 $\pm$ 0.92	+11.79$\pm$2.99	+25.47$\pm$11.86	+3.55$\pm$1.95	+4.26$\pm$3.40	+18.03$\pm$12.04	+0.61 $\pm$ 6.23
<i>Cross-backbone mean</i>	-0.55	+13.29	+8.98	+2.03	+5.27	+17.52	+4.12

Uncertainty guidance focuses corrections on structurally important regions; without it, corrections spread uniformly, yielding higher CNR (+13.5%) but lower TCI (+8.6% vs. +11.4%) and BS (+3.2% vs. +4.1%). The higher CNR without uncertainty is misleading: uniform corrections inflate the CNR numerator at the cost of reduced tissue contrast and

boundary sharpness.

No single ablation variant dominates across all metrics, confirming each component addresses a distinct aspect of the quality-fidelity balance.

Predicate threshold sensitivity. Perturbing each τ_i by $\pm 20\%$ and retraining yields CNR variation $< \pm 1.5\%$ absolute

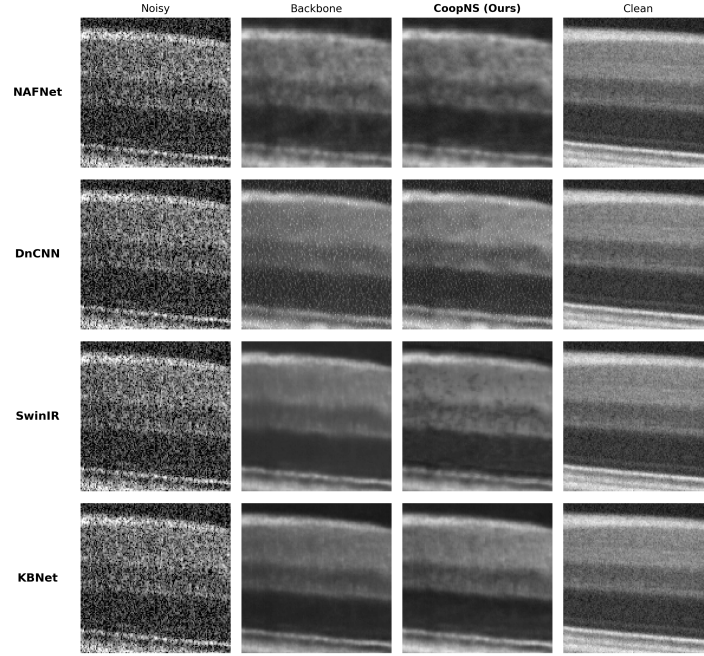


Fig. 3. Subjective quality on Duke17 subject 9 (cross-dataset, LOO). Same layout as Fig. 2. CoopNS-OCT improves tissue–background contrast on out-of-distribution data from a different scanner, demonstrating cross-domain generalization.

TABLE VII

ABLATION STUDY ON PKU37 (NAFNET BACKBONE). Δ PSNR IN DB; OTHERS IN $\Delta\%$.

Config.	Δ PSNR	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL
Full	−1.09	+12.5	+11.4	+0.3	+4.1	+29.4
w/o Negotiator	−1.01	+10.8	+13.5	+0.3	+4.1	+4.6
w/o Edge Rec.	−0.62	+7.5	+7.0	+0.7	+6.8	+2.6
w/o Uncert.	−0.94	+13.5	+8.6	+0.7	+3.2	+27.4

and PSNR within ± 0.15 dB, because learnable thresholds converge to similar operating points regardless of initialization (converged values differ by < 0.04).

E. Comparison with Alternative Enhancement Strategies

To justify the need for a learned, structure-aware correction framework, we compare CoopNS-OCT against five representative alternatives applied to the same frozen NAFNet backbone output on the PKU37 test set (173 images). All methods share the same Otsu-based tissue mask (computed from the backbone output) for fair CNR evaluation. Table VIII reports absolute PSNR and percentage change ($\Delta\%$) from the backbone-only baseline.

Histogram-based methods. Global HE [29] destroys fidelity (PSNR -20.8 dB) and *reduces* CNR by 61.2% because intensity remapping distorts the tissue–background relationship. CLAHE [30] and BF+CLAHE still degrade CNR (-2.5% / -1.5%) and speckle statistics (ENL -72.8% / -55.6%), as histogram remapping lacks spatial selectivity.

Unsharp masking [32] (USM), a standard clinical OCT enhancement, achieves the smallest PSNR penalty (-0.5 dB) and marginal CNR gain ($+0.7\%$). However, edge amplification degrades EPI (-2.6%) by boosting noise edges alongside real

TABLE VIII

COMPARISON OF POST-PROCESSING STRATEGIES ON PKU37 (NAFNET BACKBONE). PSNR IN DB (ABSOLUTE); Δ COLUMNS SHOW % CHANGE FROM BACKBONE-ONLY BASELINE. BEST Δ IN BOLD.

Method	PSNR	Δ PSNR	Δ CNR	Δ EPI	Δ ENL	Δ SNR
Backbone only	30.75	—	—	—	—	—
+ HE [29]	9.91	−20.8	−61.2	−79.7	−99.1	−84.1
+ CLAHE [30]	22.81	−7.9	−2.5	−9.3	−72.8	−48.9
+ BF+CLAHE [31]	23.26	−7.5	−1.5	−6.4	−55.6	−34.0
+ USM [32]	30.27	−0.5	+0.7	−2.6	−42.4	−22.9
+ CNR-FT	27.26	−3.5	−14.5	−0.9	−64.8	−31.5
+ CoopNS-OCT	29.66	−1.1	+12.5	+0.3	+29.4	−0.3

boundaries, and severely reduces ENL (-42.4%) and SNR (-22.9%) by amplifying speckle.

CNR-aware fine-tuning (unfreezing the backbone with composite clinical loss) *degrades* CNR by 14.5% despite a 3.5 dB PSNR loss, with catastrophic ENL reduction (-64.8%). Coupling fidelity and clinical objectives in a single backbone creates destructive gradient interference, directly motivating our decoupled two-stage approach.

CoopNS-OCT is the *only* method that improves CNR ($+12.5\%$) and EPI ($+0.3\%$) while *simultaneously* improving ENL ($+29.4\%$), with a controlled PSNR trade-off (-1.1 dB). The multiplicative gain formulation exploits OCT speckle’s multiplicative nature [2], directly increasing the CNR numerator while preserving relative structure; the additive edge module complements this by targeting boundary sharpness independently of local intensity. Every corrected pixel traces to specific predicate failures and allocation weights, supporting FDA explainability requirements for clinical AI [28].

VII. DISCUSSION

Interpretability and clinical deployment. For every corrected pixel, clinicians can inspect the confidence map (c),

predicate scores (s_1 – s_6) with spatial failure maps, compound rule activations (r_{ec} , r_{comp}), and the allocation map (**a**). This audit trail satisfies emerging regulatory transparency requirements [28]. Unlike opaque neural post-processing, CoopNS-OCT localizes failures to specific predicates and rules, enabling targeted refinement.

Generalizability. Cross-dataset LOO results confirm that the six predicates capture scanner-invariant quality attributes. This stems from two choices: (i) predicates operate on local image statistics (gradients, variances, autocorrelations) physically meaningful across scanners, and (ii) the small corrector capacity ($\sim 168K$ parameters) prevents memorizing scanner-specific artifacts. The modular design facilitates extension to other coherent imaging modalities (e.g., ultrasound, SAR) where speckle follows similar multiplicative statistics [2].

Limitations. The bounded correction range ($\epsilon = 0.20$) cannot compensate for severe backbone artifacts. The DnCNN-specific negative ΔTCI on cross-dataset evaluation reflects a beneficial effect: DnCNN’s residual speckle yields $TCI_{backbone} = 1.32 > 1.0$ on Duke17, and the corrector moves TCI toward the ideal value of 1.0 (distance from ideal decreases from 0.32 to 0.17). A distance-from-ideal metric would more transparently capture this behavior across all backbones. While our improvements exceed clinical thresholds [24], prospective reader studies remain necessary.

VIII. CONCLUSION

We presented CoopNS-OCT, a cooperative neuro-symbolic framework addressing the fundamental tension between pixel-level fidelity and clinical quality in deep OCT denoisers. Four contributions—uncertainty-guided cooperative correction, hierarchical symbolic negotiation via differentiable Lukasiewicz fuzzy logic, backbone-agnostic design, and region-aware CNR-preserving constraints—selectively improve clinical attributes while bounding fidelity degradation through formal worst-case guarantees. The pipeline adds only $\sim 168K$ parameters ($\sim 2.4\%$ overhead, $\sim 8\%$ inference time) to any frozen backbone, with every pixel-level correction traceable to specific predicate failures and negotiator allocations, providing the audit trail increasingly required for clinical AI deployment.

LOO cross-validation on Duke17 and Duke2013 with four backbone families demonstrates consistent clinical improvements (CNR +10–15%, BS +2.5–11%) with controlled PSNR trade-off (-0.25 to -0.80 dB). Three of four backbones (NAFNet, KBNNet, SwinIR) show positive improvements on *all* individual clinical metrics; the DnCNN-specific TCI degradation (-12%) is confined to a single backbone and explained by its residual architecture. CNR improvements exceed inter-reader variability [24] by 1.5–3 $\times$, with effect sizes (Cohen’s $d > 2.5$) and statistical significance ($p < 10^{-30}$ sign test across 136 test cases) confirming reliability despite modest dataset sizes. Comparison against five alternative strategies confirms that selective, structure-aware correction outperforms global enhancement. Future work will extend to 3D volumetric OCT with inter-slice consistency predicates, explore backbone-adaptive predicate weighting, and conduct prospective reader studies.

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